

BIRD SPECIES OBSERVATION ANALYSIS

Ecosystem Diversity, Microclimate Dynamics & Multi-Tier Analytics Architecture

National Capital Region National Parks — 2018 Breeding Season

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Architecture: Google Colab -> SQL Server -> Streamlit

Live Dashboard: <https://bird-species-observation-analysis-5adppp8jrh6xjqws3uwfey.streamlit.app/>
Dataset Scope: 17,077 Records across 11 Parks

Target Engine: Microsoft SQL Server 2022 (BirdMonitoringDB) Temporal Scope: May 7 – July 19, 2018 (Single-Year Breeding Horizon)

1. EXECUTIVE SUMMARY & PROJECT CHARTER

1.1 Problem Statement & Scope: Understanding the distribution, diversity, and behavioral ecology of avian populations across differing ecosystem types is critical for informing biodiversity conservation, land-use planning, and ecosystem management. Avian species serve as sensitive bio-indicators whose population structures, detection frequencies, and vocalization patterns reflect underlying microclimate variations, vegetation structure, and anthropogenic disturbances.

1.2 Ecological Regime Focus: This project delivers an end-to-end data analytics, statistical modeling, and interactive web platform that evaluates avian observational data across two distinct ecosystems: (1) Forest Ecosystem (multi-layered canopy, dense understory, lower solar radiation, higher relative humidity); and (2) Grassland Ecosystem (open meadows, herbaceous vegetation, high direct solar exposure, elevated ambient temperatures).

1.3 Multi-Tier Architecture: The project was implemented across three production-ready engineering tiers: Tier 1: Google Colab & Python for raw Excel ingestion, deduplication auditing, data harmonization, feature engineering, and ecological diversity indexing; Tier 2: Microsoft SQL Server (BirdMonitoringDB) for relational persistence, non-clustered index optimization, analytical view modeling, stored procedures, and business query benchmarking; and Tier 3: Streamlit + Plotly for an 8-page public cloud dashboard deployed at 24/7 availability.

2. RAW DATASET ARCHITECTURE & INGESTION ANALYSIS (PHASE 1)

2.1 Source Workbooks: The raw observational data was provided in two Microsoft Excel workbooks containing 11 separate administrative unit sheets corresponding to National Park Service properties: Bird_Monitoring_Data_FOREST.XLSX (~955 KB) and Bird_Monitoring_Data_GRASSLAND.XLSX (~958 KB). Each sheet was inspected programmatically using Python (pandas, openpyxl).

Admin Unit	Park Name / Description	Forest Rows	Forest Cols	Grassland Rows	Grassland Cols	Status
ANTI	Antietam National Battlefield	333	29	3,588	29	Active in Both
CATO	Catoctin Mountain Park	805	29	0	29	Forest Only
CHOH	Chesapeake & Ohio Canal NHP	2,202	29	0	29	Forest Only
GWMP	George Washington Memorial Pkwy	386	29	0	29	Forest Only
HAFE	Harpers Ferry NHP	422	29	117	29	Active in Both

MANA	Manassas National Battlefield	465	29	1,811	29	Active in Both
MONO	Monocacy National Battlefield	370	29	3,015	29	Active in Both
NACE	National Capital Parks - East	684	29	0	29	Forest Only
PRWI	Prince William Forest Park	2,463	29	0	29	Forest Only
ROCR	Rock Creek Park	289	29	0	29	Forest Only
WOTR	Wolf Trap National Park	127	29	0	29	Forest Only
TOTALS	11 Administrative Units	8,546	29	8,531	29	17,077 Combined Rows

3. DATA PREPROCESSING, CLEANING & HARMONIZATION (PHASES 2–4)

3.1 Point-Count Deduplication: In point-count surveys (National Park Service standard protocol), an observer conducts a 10-minute count at a specific plot. If multiple individuals of the same species (e.g., 3 Common Grackles) are recorded in the same interval, distance band, and plot, they legitimately share session attributes. Grassland contained 1,705 identical-attribute rows, while Forest had 0. All 17,077 rows were retained to preserve true biological abundance while generating an audit report.

3.2 Missing Value Remediation: Missingness was systematically handled without deleting valid biological records. Sex missing in 30.35% of rows (monomorphic species unable to be sexed in the field) was standardized to 'Undetermined'. Distance missing in 8.70% of rows (flyovers) was mapped to 'Not Recorded / Flyover'. Sub_Unit_Code (95.77% null) was preserved as nullable without synthesizing data.

3.3 Type & Range Validation: Dates were cast to datetime64[ns], Start/End times standardized to HH:MM:SS, taxonomic identifiers cast to nullable Int64, status flags cast to boolean, and weather metrics verified within physical atmospheric limits (Temperature: 11.0°C to 37.3°C; Humidity: 7.3% to 98.8%). All extreme values represented valid diurnal weather fluctuations and were retained.

4. FEATURE ENGINEERING & DERIVED METRICS (PHASE 5)

4.1 Engineered Dimensions: To power multi-dimensional querying, 9 analytical features were engineered: Month, Month_Name, Day, Day_Of_Week, Observation_Hour (5 to 11 AM), Observation_Duration_Min (10.0 min median), Season (Early Breeding Spring vs Peak Breeding Summer), Distance_Category (Near <=50m, Far 50-100m, Very Far >100m, Flyover), and Conservation_Priority (High Priority, Watchlist Only, Regional Stewardship Only, Standard/Secure).

5. MATHEMATICAL ECOLOGICAL DIVERSITY MODELING (PHASE 8)

5.1 Diversity Formulations: Avian community structure was modeled using formal ecological formulations: Species Richness S, Shannon-Wiener Diversity H', Simpson's Diversity 1-D, and Pielou's Species Evenness J'.

Ecosystem Habitat	Total Obs	Richness (S)	Shannon (H')	Simpson (1-D)	Evenness (J')	Mean Temp (°C)
Forest Ecosystem	8,546 (50.05%)	108	3.5780	0.9597	0.7642	21.87°C

Grassland Ecosystem	8,531 (49.95%)	107	3.7037	0.9662	0.7926	23.27°C
Overall Landscape	17,077 (100.0%)	127	3.9214	0.9715	0.8095	22.57°C

6. SPECIES SPECIALIZATION & COMMUNITY DYNAMICS (PHASE 7)

6.1 Specialization Breakdown: Of the 127 total species detected, 88 species (69.3%) were generalists occupying both ecosystems (e.g., American Robin, Northern Cardinal, Red-eyed Vireo). 20 species were strict Forest specialists (e.g., Wood Thrush, Ovenbird, Pileated Woodpecker, Scarlet Tanager), and 19 species were strict Grassland specialists (e.g., Grasshopper Sparrow, Eastern Meadowlark, Bobolink, Dickcissel).

6.2 Dominant Taxa & Modalities: Top recorded species across the landscape were American Robin (1,061 obs), Red-eyed Vireo (972 obs), Northern Cardinal (969 obs), Eastern Towhee (873 obs), Indigo Bunting (823 obs), and Wood Thrush (809 obs). Auditory detections accounted for 79.8% of all encounters (Singing: 53.3%, Calling: 26.5%, Visual: 20.2%).

6.3 Temporal Dynamics: Survey activity was strictly confined to the 2018 breeding season (May 7 – July 19). Observations peaked in June (6,596 obs, 38.6%) during peak territorial vocalization. Daily detections peaked sharply during the dawn chorus window between 06:00 AM and 08:00 AM (51.6% of all records).

7. CONSERVATION & AT-RISK POPULATION ANALYSIS

7.1 Conservation Status: Avian populations were evaluated against Partners in Flight (PIF) Watchlist and Regional Stewardship criteria: 2,429 observations represent PIF Watchlist species of continental concern (Wood Thrush, Prairie Warbler, Grasshopper Sparrow, Kentucky Warbler, Cerulean Warbler). 2,933 observations represent species of high Regional Stewardship responsibility (Eastern Towhee, Eastern Wood-Pewee, Field Sparrow, Scarlet Tanager).

Conservation Priority Tier	Total Observations	Unique Species	Plots Recorded	Key Indicator Species
High Priority (Watchlist & Stewardship)	367	4	225	Wood Thrush, Prairie Warbler, Grasshopper Sparrow
PIF Watchlist Only	11	3	11	Cerulean Warbler, Golden-winged Warbler
Regional Stewardship Only	3,618	20	598	Eastern Towhee, Eastern Wood-Pewee, Field Sparrow
Standard / Secure	13,081	100	609	American Robin, Northern Cardinal, Red-eyed Vireo

8. MICROSOFT SQL SERVER DATABASE ARCHITECTURE (PHASE 12)

8.1 Database Implementation: A relational database BirdMonitoringDB was engineered in SQL Server with table dbo.Bird_Observations (43 columns, Primary Key Observation_ID). Five non-clustered indexes were created on Habitat, Common_Name, Admin_Unit_Code, Date, and Conservation_Priority. Six analytical reporting views (vw_Habitat_Summary, vw_Species_Distribution, vw_Temporal_Trends, vw_Spatial_Hotspots, vw_Conservation_Priorities, vw_Behavior_Detection) and three stored procedures were deployed to support modular reporting.

9. STREAMLIT + PLOTLY INTERACTIVE WEB APPLICATION (PHASE 3)

9.1 Web Dashboard Architecture: The interactive dashboard is publicly hosted at <https://bird-species-observation-analysis-5adppp8jrh6xjqws3uwfey.streamlit.app/>. The application features 8 analytical modules: (1) Executive Overview, (2) Habitat & Ecosystems, (3) Species & Biodiversity, (4) Temporal & Diurnal Dynamics, (5) Spatial & Hotspot Explorer, (6) Environmental & Behavior, (7) Conservation Center, and (8) Data Explorer & SQL Query Console.

10. STRATEGIC RECOMMENDATIONS & POLICY GUIDANCE

10.1 Key Recommendations:

1. Forest Canopy Protection: Mature interior forests in Prince William Forest Park (PRWI) and C&O Canal (CHOH) must be protected from fragmentation to safeguard Wood Thrush nesting territories.
2. Grassland Mowing Regimes: Implementing delayed mowing schedules until mid-July at Antietam (ANTI) and Monocacy (MONO) will protect ground-nesting Grasshopper Sparrows and Eastern Meadowlarks.
3. Eco-Tourism Development: High-richness plots (ANTI-0105, MONO-0057, CHOH-0812) should be incorporated into guided eco-tourism birding trails.
4. Standardized Monitoring Expansion: Expanding Grassland point counts to units currently lacking grassland data (CATO, GWMP, PRWI) will provide a complete landscape-level monitoring network.